

# YOUNGRAE KIM

youngrae@usc.edu   youngraekimm.github.io

## EDUCATION

### University of Southern California (USC)

Ph.D. in Computer Engineering (Advisor: Prof. Jay Kuo, Prof. Peter Beerel)  
– Cumulative GPA: 3.83 / 4.0

Los Angeles, CA, USA  
Aug. 2024 – Present

### Korea Advanced Institute of Science and Technology (KAIST)

M.S. in Computer Science (Advisor: Prof. Dongman Lee)  
– Thesis: "Few-Shot Weather-Degraded Image Restoration"  
– Cumulative GPA: 3.62 / 4.0

Daejeon, Korea  
Feb.2022 – Feb.2024

### Hongik University

B.S. in Computer Engineering (Advisor: Prof. Young Yoon)  
– Major GPA: 98.41%; Cumulative GPA: 95.10%

Seoul, Korea  
Mar.2016 – Feb.2022

## RESEARCH INTERESTS

My research interests lie in **video generation**, **world models**, and **3D visual perception**, with a focus on building temporally coherent, 3D-aware generative systems. I am also interested in **video understanding** with **multimodal LLMs**.

## RESEARCH EXPERIENCE

### Google

*Student Researcher*, Host: Srinivas Kaza, Xuan Luo

May. 2026 – Present

### USC

*Research Assistant*, Advisor: Prof. Jay Kuo, Prof. Peter Beerel

Aug. 2024 – Present

### KAIST

*Research Assistant*, Advisor: Prof. Dongman Lee, Prof. Seunghoon Hong

Feb. 2022 – Jul. 2024

## PUBLICATIONS

- P1. **EVIDENT: Routing MLLM Adaptation through Entity-Grounded Visual Evidence for Cross-Domain Video Temporal Grounding**  
Geo Ahn\*, Jiwook Han\*, **Youngrae Kim\***, Joonseok Lee, Jinwoo Choi  
*Pre-Print*
- C11. **Locality-Aware Redundancy Pruning for LLM Depth Compression**  
Vincent-Daniel Yun\*, **Youngrae Kim\***, Woosang Lim\*, YoungJin Heo, Minkyu Kim, Sunwoo Lee  
*Conference on Empirical Methods in Natural Language Processing (EMNLP) 2026*
- C10. **Perturbation-Aware Training and Hardware Chopping for Robust Analog Neural Network Accelerators**  
Maysara Hamada\*, **Youngrae Kim\***, Sreetama Sarkar, Chee-An Yu, Jake Hyun, Huixuan Yin, Hsiang-Chun Cheng, Jian Meng, Yifan He, Jae-sun Seo, C.-C. Jay Kuo, Peter Beerel, and Chen Shuo-Wei  
*International Conference on Computer-Aided Design (ICCAD) 2026*
- C9. **MemRoPE: Training-Free Infinite Video Generation via Evolving Memory Tokens**  
**Youngrae Kim\***, Qixin Hu\*, C.-C. Jay Kuo, Peter Beerel  
*European Conference on Computer Vision (ECCV) 2026*
- C8. **Rethinking Layer Redundancy in Large Language Models: Calibration Objectives and Search for Depth Pruning**  
Minkyu Kim\*, Vincent-Daniel Yun\*, **Youngrae Kim\***, YoungJin Heo\*, Suin Cho\*, Seong-hun Kim\*, Woosang Lim\*, Gaeul Kwon\*  
*Resource-Adaptive Foundation Model Inference Workshop at International Conference on Machine Learning (ICML) 2026*
- C7. **Beyond the Mean: Three-Axis Fidelity for Aligning LLM-Based Survey Simulators from Small Pilot Data**  
Eun Cheol Choi\*, **Youngrae Kim\***, Prabhu Pugalenth\*, Hong-En Chen\*, Bo-Ruei Huang\*  
*Pluralistic Alignment Workshop at International Conference on Machine Learning (ICML) 2026*

- C6. **SlotVTG: Object-Centric Adapter for Generalizable Video Temporal Grounding**  
 Jiwook Han\*, Geo Ahn\*, **Youngrae Kim\***, Jinwoo Choi  
*GRAIL-V Workshop at IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026*
- C5. **Feature Augmentation based Test-Time Adaptation**  
 Yeonggeol Cho\*, **Youngrae Kim\***, Junho Yoon, Seunghoon Hong, Dongman Lee  
*IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2025*
- C4. **DEVIAS: Learning Disentangled Video Representations of Action and Scene**  
**Youngrae Kim\***, Kyungho Bae\*, Geo Ahn\*, Jinwoo Choi  
*European Conference on Computer Vision (ECCV) 2024 – (Oral Presentation, 2.3% acceptance rate)*
- C3. **MetaWeather: Few-Shot Weather-Degraded Image Restoration**  
**Youngrae Kim\***, Yeonggeol Cho\*, Thanh-Tung Nguyen, Seunghoon Hong, Dongman Lee  
*European Conference on Computer Vision (ECCV) 2024*
- C2. **Beyond Entropy: Style Transfer Guided Single Image Continual Test-Time Adaptation**  
 Yeonggeol Cho\*, **Youngrae Kim\***, Dongman Lee  
*Workshop on Vision-Centric Autonomous Driving at European Conference on Computer Vision (ECCV) 2024*
- C1. **Efficient Reference-based Video Super-Resolution (ERVSR): Single Reference Image is All You Need**  
**Youngrae Kim\***, Jinsu Lim\*, Hoonhee Cho\*, Minji Lee\*, Dongman Lee, Kuk-Jin Yoon, Ho-Jin Choi  
*IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2023 – (Oral Presentation)*
- J1. **Zone-Agnostic Greedy Taxi Dispatch Algorithm Based on Contextual Matching Matrix for Efficient Maximization of Revenue and Profit**  
**Youngrae Kim**, Young Yoon  
*Electronics 2021*  
 (\* denotes equal contribution)

## PATENTS

- Dongman Lee, Yeonggeol Cho, **Youngrae Kim**, Junho Yoon. Korean Patent, Application No. 10-2025-0066479. Single Image Based Style Transfer Guided Continual Test-Time Adaptation System. 2025.

## FELLOWSHIPS AND HONORS

|                                                                                                    |             |
|----------------------------------------------------------------------------------------------------|-------------|
| Outstanding Poster Presentation Award, Workshop on Image Processing and Image Understanding (IPIU) | Feb. 2026   |
| Oral Presentation, European Conference on Computer Vision (ECCV)                                   | Sep. 2024   |
| Best Paper Award, Korea Computer Congress                                                          | Jun. 2024   |
| Honorable Mention, Workshop on Image Processing and Image Understanding (IPIU)                     | Feb. 2024   |
| Full-tuition Government Scholarship for Science and Engineering, KAIST                             | 2022 – 2024 |
| Oral Presentation, IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)            | Jan. 2023   |
| Best TA Award, KAIST                                                                               | Fall 2022   |
| Academic Excellence Scholarship for 7 semesters, Hongik University                                 | 2016 – 2022 |

## ACADEMIC SERVICES

*Invited Talk: Toward Infinite Video Generation, eBay (host: Ashirbad Mishra),* May 2026  
*Reviewer: CVPR 2024, 2025, 2026; ECCV 2026; NeurIPS 2026; AAAI 2025, 2026; WACV 2025; BMVC 2026; TCSVT*

## TEACHING

|                                                          |             |
|----------------------------------------------------------|-------------|
| <i>Teaching Assistant, CS206 Data Structure, KAIST</i>   | Fall 2022   |
| <i>Teaching Assistant, CS330 Operating System, KAIST</i> | Spring 2022 |

## LEADERSHIP EXPERIENCE

|                                                                                  |                       |
|----------------------------------------------------------------------------------|-----------------------|
| <i>Vice President, Korean Graduate Student Association (KGSA), USC</i>           | May 2026 – Present    |
| <i>Troop Leader (Sergeant, Honorable Discharge), Republic of Korea Air Force</i> | Apr. 2018 – Mar. 2020 |